

Statistical analysis of experimental data

Introduction to Python

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Lecture 02
October 9, 2025

Lecture concept

The goal of the lecture is not only to present the theoretical concepts but also to demonstrate how they can be applied in practical terms to data analysis.

Different concepts as well as analysis methods will be discussed using code examples.

We will address many (relatively simple) problems and try to look at their numerical solutions.

All examples presented are based on Python and multiple additional packages, which will be shortly introduced today.

Web resources

Kampus platform will be used for lecture slides, Python notebooks, as well as home exercises (and final exam):

<https://kampus-kursy.ckc.uw.edu.pl/course/view.php?id=5398>

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All scripts discussed at the lecture will be available for download from the github repository:

<https://github.com/zarnecki/SAED>

Inserted in the lecture slides are links to open them in your browser using the Google Colab.

You can also use this link  [Open in Colab](#) to open repository and select the notebook to load.

Please remember that you need to save the file to (your) Google Disc before you make any modifications. Otherwise, all your changes will be lost!..

JupyterLab

All scripts discussed during the course were developed with [Jupyter Lab](#)

Jupyter's tools are available for installation via the Python Package Index (pip)

Containers

If you want to avoid installing and configuring all the required packages on your personal computer, you can also use virtual environments known as containers (**lightweight versions of old time virtual machines**). Two popular environments used to run the containers:

- [Apptainer](#)
- [Docker](#)

Both Docker and Apptainer can use the same input files.

Docker is available for any operating system, Apptainer only for Linux ones.

Notebooks presented at this lecture can be run with [akalinow/root-fedora39](#) container prepared for the other course some time ago. This is a container based on Fedora Linux ditribution, and contains all packages we will use during the classes.

Python tools

One of the main advantages of Python is the large number of diverse packages developed for various applications...

During the course we will use the following packages for computation and plotting:

- [numpy](#) - The fundamental package for scientific computing with Python
- [matplotlib](#) - Visualization with Python
- [SciPy](#) - Fundamental algorithms for scientific computing in Python
- [scikit-learn](#) - Machine Learning in Python, simple and efficient
- [pandas](#) - Handling more complex data structures

See lecture notebook for more details and examples:





NumPy documentation

Version: 2.0.dev0

Download documentation: [Historical versions of documentation](#)

Useful links: [Installation](#) | [Source Repository](#) | [Issue Tracker](#) | [Q&A Support](#) | [Mailing List](#)

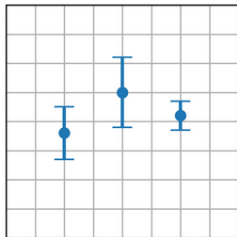
NumPy is the fundamental package for scientific computing in Python. It is a Python library that provides a multidimensional array object, various derived objects (such as masked arrays and matrices), and an assortment of routines for fast operations on arrays, including mathematical, logical, shape manipulation, sorting, selecting, I/O, discrete Fourier transforms, basic linear algebra, basic statistical operations, random simulation and much more.



Getting Started



User Guide



`errorbar(x, y, yerr, xerr)`

Matplotlib: Visualization with Python

Matplotlib is a comprehensive library for creating static, animated, and interactive visualizations in Python. Matplotlib makes easy things easy and hard things possible.

- Create [publication quality plots](#).
- Make [interactive figures](#) that can zoom, pan, update.
- Customize [visual style and layout](#).
- Export to [many file formats](#).
- Embed in [JupyterLab](#) and [Graphical User Interfaces](#).
- Use a rich array of [third-party packages](#) built on Matplotlib.

Try Matplotlib (on Binder)



Getting Started



Examples



Reference



Cheat Sheets



Documentation

[Install](#) [Documentation](#) [Community](#) [About Us](#) [Contribute](#)



Fundamental algorithms for scientific computing in Python

GET STARTED

SciPy 1.11.3 released! 2023-09-27

FUNDAMENTAL ALGORITHMS

SciPy provides algorithms for optimization, integration, interpolation, eigenvalue problems, algebraic equations, differential equations, statistics and many other classes of problems.

BROADLY APPLICABLE

The algorithms and data structures provided by SciPy are broadly applicable across domains.

pandas documentation

Date: Sep 20, 2023 **Version:** 2.1.1

Download documentation: [Zipped HTML](#)

Previous versions: Documentation of previous pandas versions is available at pandas.pydata.org.

Useful links: [Binary Installers](#) | [Source Repository](#) | [Issues & Ideas](#) | [Q&A Support](#) | [Mailing List](#)

pandas is an open source, BSD-licensed library providing high-performance, easy-to-use data structures and data analysis tools for the [Python](#) programming language.



Getting started

New to *pandas*? Check out the getting started guides.



User guide

The user guide provides in-depth information on the



[Install](#) [User Guide](#) [API](#) [Examples](#) [Community](#) [More ▾](#)

 [Go](#)

scikit-learn

Machine Learning in Python

[Getting Started](#)[Release Highlights for 1.3](#)[GitHub](#)

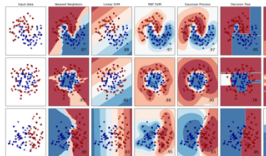
- Simple and efficient tools for predictive data analysis
- Accessible to everybody, and reusable in various contexts
- Built on NumPy, SciPy, and matplotlib
- Open source, commercially usable - BSD license

Classification

Identifying which category an object belongs to.

Applications: Spam detection, image recognition.

Algorithms: Gradient boosting, nearest neighbors, random forest, logistic regression, and more...



Examples

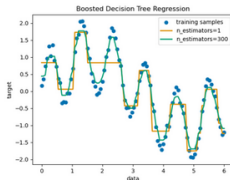
A.F. Żarnecki

Regression

Predicting a continuous-valued attribute associated with an object.

Applications: Drug response, Stock prices.

Algorithms: Gradient boosting, nearest neighbors, random forest, ridge, and more...



Examples

Statistical analysis 02

Clustering

Automatic grouping of similar objects into sets.

Applications: Customer segmentation, Grouping experiment outcomes

Algorithms: k-Means, HDBSCAN, hierarchical clustering, and more...



Examples

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Graphical presentation of the data

Look for the recent data describing age and gender structure of the population of Poland. One of the possible sources is the [Demographic Database](#) of the [Central Statistical Office](#) (Polish: [Główny Urząd Statystyczny](#)) or [Poland's Data Portal](#).

- 1 Prepare the plot showing the gender balance (men to women ratio) as a function of age.
- 2 Can the observed dependence be used to draw any conclusions concerning the life expectancy? What your conclusion would be? Justify your answer.

Solutions should be uploaded until October 22.

Solutions should be uploaded to Kampus as PDF documents including description/justification of the approach used, numerical results and relevant plots, as well as result discussion.

The Python notebook can be uploaded as a second file
or the link to notebook in Google Colab can be added as a comment.

Make sure your code runs “from scratch” (no hidden dependencies) and Google Colab accessible.